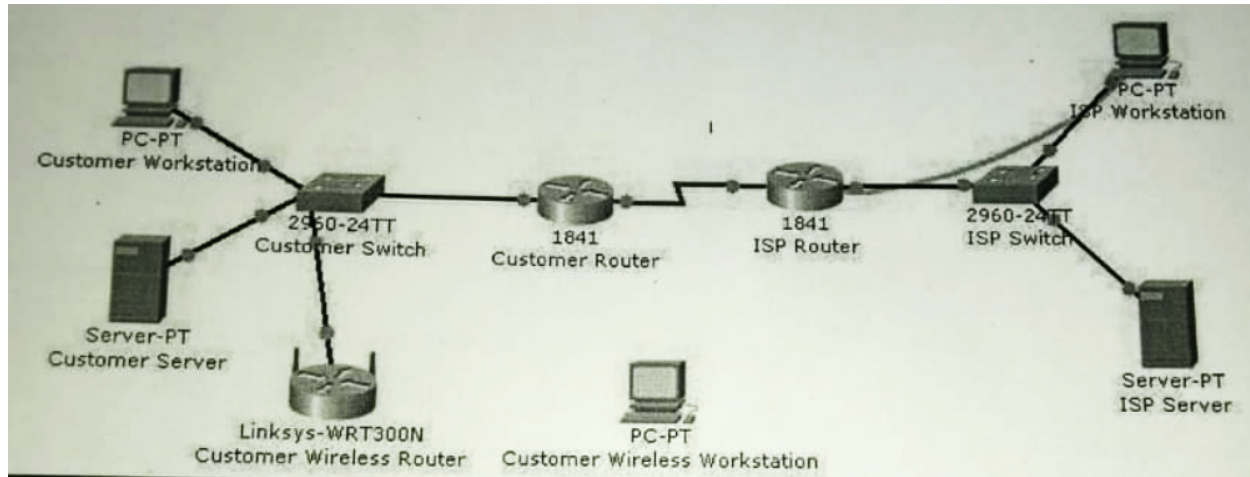


Question #1:

You have been asked to go back to a business customer and install a new Linksys wireless router for the customer office. The company has some new personnel who will be using wireless computers to save money on adding additional wired connections to the building. The business is concerned about the security of the network because they have financial and highly classified data being transmitted over the network. Your job is to configure the security on the router to protect the data. You need to configure WEP security on both a Linksys wireless router and a workstation.

**Answer:****Step-by-Step Configuration****1. Configure WEP on Linksys Router**

1. Open the **Linksys-WRT300N (Customer Wireless Router)** in Packet Tracer.
2. Go to the **GUI tab** where most of the router's web-based settings are available.
3. Click on the **Wireless** menu, then choose the **Wireless Security** sub-tab.
4. Under **Security Mode**, change the setting from *Disabled* to **WEP**.
5. Now enter the security details carefully:
 - **Security Mode:** WEP
 - **Default Transmit Key:** 1
 - **WEP Encryption:** 128-bit (26 hex digits) for stronger security
 - **Key 1 (WEP key):** Example → ABCDE12345ABCDE12345ABCDE12
6. Scroll down and click **Save Settings** to apply the configuration.

At this point, the wireless router will start broadcasting its SSID with WEP protection enabled.

2. Configure Wireless Workstation (PC-PT)

1. Open the **Customer Wireless Workstation (PC-PT)**.
2. Go to the **Desktop tab** → **PC Wireless**. This is where the wireless network card settings are found.
3. The system will scan for available wireless networks. Select the **SSID** of the Linksys router that you configured.
4. When prompted for security, choose **WEP** as the security type.

5. Enter the exact same **WEP Key** that was configured on the router. For example:
`ABCDE12345ABCDE12345ABCDE12`
6. Click **Connect**. The wireless PC should now connect to the router securely.

3. Verify the Connection

1. On the **Wireless PC**, open the **Command Prompt** from the Desktop tab.
2. First, test connectivity to the router by typing:
`ping 192.168.1.1`
3. Next, test communication with other devices in the LAN such as the **Customer Server** or **Customer Workstation** by using their IP addresses:
`ping 192.168.1.100`
If the ping is successful, it confirms that the wireless workstation is securely connected and can access other resources on the network.

Result: The wireless connection is now secured with **WEP**, ensuring that only users who know the WEP key can connect to the customer's wireless network.

Final Note:

- WEP is outdated and not recommended in real life, but in **Packet Tracer labs** this is usually required.
- Always make sure the **SSID**, **WEP key**, and **encryption type** match on both router and workstation.

Question #2:

Suppose you have a class C Network 206.94.115.0. Your task is to design a subnet scheme so that we can create 14 Network segments (subnets) within this Network. Each subnet should support 10-12 hosts.

- a. How many bits would you use for the subnet ID?
- b. How many bits would you use for the Host ID?
- c. How many maximum possible subnets will be there?
- d. How many maximum possible hosts will be there in each subnet?
- e. Write down the Subnet Mask of your scheme.

Write down All the valid IP addresses, the broadcast address of the First subnet of your scheme.

Answer:

Subnetting Solution

We are given a **Class C network: 206.94.115.0/24**.

Our task is to create **14 subnets**, where each subnet supports about **10–12 hosts**.

(a)

Step 1: Determine Number of Subnets

- A Class C network has **8 host bits** available (since default mask is /24).
- To create at least 14 subnets, we need to **borrow subnet bits** from the host portion.
- Formula: $2^n \geq \text{required subnets}$
- Here, $2^n \geq 14 \rightarrow n = 4$ (since $2^4 = 16$).

So we borrow **4 bits** for the subnet ID.

(b)

Step 2: Determine Number of Host Bits

- Total host bits in Class C = 8.
- Borrowed = 4.
- Remaining for hosts = $8 - 4 = 4$ bits.
- Formula: $2^h - 2 = \text{usable hosts}$
- Here, $2^4 - 2 = 14$ hosts.

So each subnet can support up to **14 hosts**, which is enough for the requirement (10–12).

(c)

Step 3: Maximum Numbers

- **Maximum subnets:** $2^4 = 16$
- **Maximum usable hosts per subnet:** 14

(d)

Step 4: Subnet Mask

- Default Class C mask = 255.255.255.0 (/24).
- Borrowed 4 bits → New mask = /28.
- Subnet mask = **255.255.255.240**

(e)

Step 5: Calculate Subnet Increments

- Block size = $256 - 240 = 16$.
- So the subnets will increase by 16 in the last octet.
- Subnets: 206.94.115.0, 206.94.115.16, 206.94.115.32, 206.94.115.48 ... up to 206.94.115.240.

(f)

Step 6: First Subnet Details

- **Network ID:** 206.94.115.0
- **Valid Host Range:** 206.94.115.1 → 206.94.115.14
- **Broadcast Address:** 206.94.115.15

Result:

Bits for Subnet ID: 4

Bits for Host ID: 4

Maximum Subnets: 16

Maximum Hosts per Subnet: 14

Subnet Mask: 255.255.255.240 (/28)

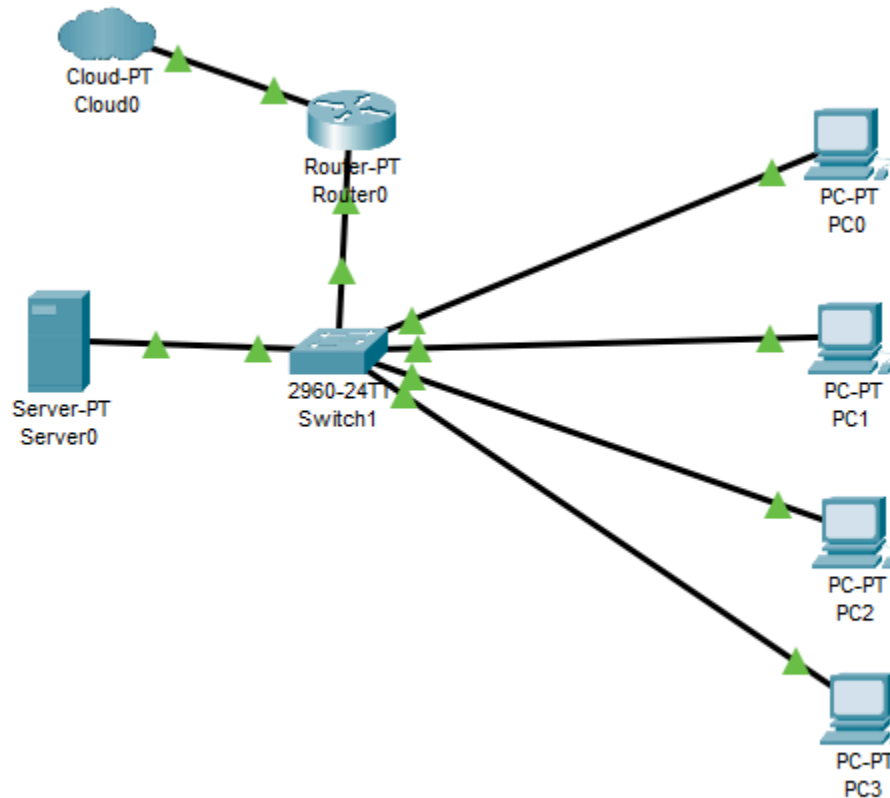
First Subnet:

- Network Address: 206.94.115.0
- Usable IPs: 206.94.115.1 – 206.94.115.14
- Broadcast: 206.94.115.15

Question#3:

Create a mini project: Design and simulate a small office network with DHCP, DNS, and NAT enabled. Use Packet Tracer or similar tools.

Answer:



To design and simulate a small office network with DHCP, DNS, and NAT, the following steps were performed in Cisco Packet Tracer:

Step 1: Network Topology

- Devices used: 1 Router-PT, 1 Switch-2960, 4 PCs, 1 Server, and 1 Cloud (Internet).
- PCs and Server connected to Switch.
- Switch connected to Router Fa0/0.
- Router Fa1/0 connected to Cloud.

Step 2: DHCP and DNS Server Configuration

- DHCP Pool Name: *office*
- Default Gateway: 192.168.1.1
- DNS Server: 192.168.1.2
- Starting IP Address: 192.168.1.100
- Subnet Mask: 255.255.255.0
- Maximum Users: 50
- Enabled DNS, HTTP, and HTTPS services.

Step 3: Router Configuration (NAT Setup)

On the router CLI:

```
Router> enable
```

```
Router# configure terminal
```

! Assign IP addresses

```
Router(config)# interface fastEthernet0/0
```

```
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

```
Router(config)# interface fastEthernet1/0
Router(config-if)# ip address 200.1.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

! Define inside/outside

```
Router(config)# interface fastEthernet0/0
Router(config-if)# ip nat inside
Router(config-if)# exit
```

```
Router(config)# interface fastEthernet1/0
Router(config-if)# ip nat outside
Router(config-if)# exit
```

! NAT Configuration

```
Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)# ip nat inside source list 1 interface fastEthernet1/0
overload
```

Step 4: Testing

- PCs automatically received IPs via DHCP.
 - Successful ping to Gateway (192.168.1.1) and Server (192.168.1.2).
 - DNS resolved domain names (e.g., *www.office.com*).
 - HTTP/HTTPS access to the server was successful.
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